Binding Pocket Residue Prediction Challenge

Student Number: k51843919

Student Name: Lukas Steu

In this task, the goal was to predict which amino acid residues of a protein are part of a binding pocket based on the provided PDB structure files. The training data consisted of PDB files together with residue-level annotations from train.csv, while the test set only contained the structures. The required submission format was one row per protein with a space-separated list of predicted residue IDs. The official evaluation metric was the Jaccard score aka the IoU (Intersection of Units).

As a first step, I parsed all PDB files with Biopython (Stackoverflow helped me out there). I only kept standard amino acids that had an alpha carbon atom, because my feature extraction was based on residue identity and alpha carbon coordinates. To make the labels and parsed residues match correctly, I converted each residue into the same format as used in the CSV file. Based on this, I created a residue-level training dataframe where each row corresponds to one residue and the target is 1 if the residue is part of the binding pocket and 0 otherwise. During preprocessing I noticed that a few labelled residues could not be found in the parsed structures, so for training I only used the residues that were actually present in both the labels and the PDB file.

For the features, I combined simple biochemical information with geometric features from the protein structure. The biochemical features included the amino acid itself and whether it is hydrophobic, polar, positively charged, negatively charged, or aromatic. In addition, I computed several geometry-based features from the alpha carbon coordinates, such as the distance to the protein center, the z-scored distance to the center, the number of neighboring residues within different radii, nearest-neighbor distance statistics, and the local composition of surrounding residues. These recommendations for the geometric features were made by ChatGPT, which explained that the motivation behind these features was that binding pockets are often characterized by a specific local chemical environment and by residues being located in certain structural regions.

For the model, I used a LightGBM classifier on the residue-level feature table (because this actually runs on my 9-year-old CPU). Since residues from the same protein are highly related, I used GroupKFold cross-validation with pdb_id as grouping variable to make sure that residues from the same protein never appear in both training and validation data. The final model used n_estimators=800, learning_rate=0.03, num_leaves=127, min_child_samples=50, subsample=0.8, colsample_bytree=0.8, reg_alpha=0.5, reg_lambda=1.0 and class_weight="balanced". The categorical features were aa and chain_id. For the final test predictions, I averaged the predicted probabilities over all fold models.

Because the model outputs probabilities for each residue, I still had to decide which residues should finally be classified as pocket residues. For this, I optimized a threshold on the out-of-fold predictions using a local evaluation that follows the official Jaccard-based metric. As an additional post-processing step, I kept only the largest spatially connected component among the predicted residues. The idea behind this was that real binding pockets usually form one compact region, while isolated predicted residues are often false positives. If no large enough connected component was found, I kept the highest-scoring residue as a fallback.